Grid Computing for Belle II Analysis

Certificates · VOMS · gbasf2 — hands-on with the new IHEP CA

Xiao Han · on behalf of Grid Group, IHEP Computing Center
hanx@ihep.ac.cn

November 14–18, 2026 · Southeast University, Nanjing

Winter School 2026 for Belle II China Group, gridca.ihep.ac.cn

Outline

- **The Grid & the New CA**
  - *What the grid is · what changed*
- **Before You Start**
  - *Preparation · certificate types*
- **Hands-on: User Certificate**
  - *Apply · review · approve · download*
- **Migrate from IHEP-2013**
  - *One-time path for old certificates*
- **Use It on the Grid**
  - *VO · VOMS · proxy · gbasf2*
- **Renewal & Revocation**
  - *Keep it valid · revoke when compromised*
- **Robot & Host Certificates**
  - *For services and machines*
- **Troubleshooting & Security**
  - *FAQ · safe habits · security*

1 · The Grid & the New CA

What is the Belle II grid?

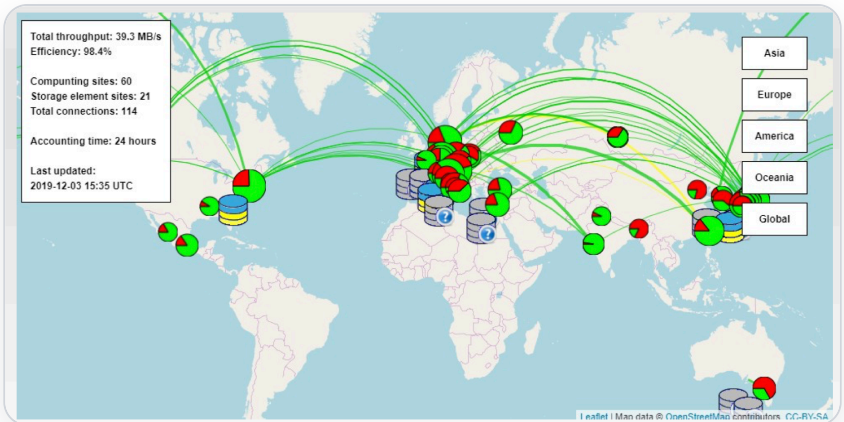
> **Note**
> The grid is a **distributed computing system** used by Belle II — and many other experiments — to harness computing resources of universities and institutes worldwide.

Why not one big data center?

- Modern experiments collect **tens of petabytes**
- Processing all data / MC at a single site is **not sustainable**
- Belle II runs on grid sites across Asia, Europe and America

The grid is used for:

- MC production · data reprocessing · skimming
- **Your analysis** — this is where you come in



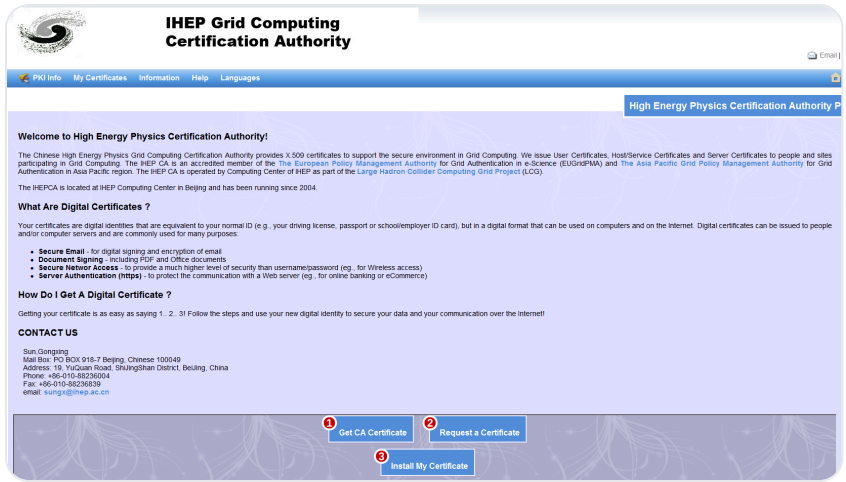
Grid sites and transfers worldwide — computing centers, storage elements and their connections.

View the live map at belle2.jp/computing (opens in your browser).

Your path today:

certificate → VO membership → proxy → gbasf2 jobs

What changed: OpenCA → OpenXPKI



cagrid.ihep.ac.cn — the old CA web interface

Dimension	Old (OpenCA)	New (OpenXPKI)
URL	cagrid.ihep.ac.cn	**gridca.ihep.ac.cn**
Login	Basic web form	**IHEP Account** + cert migration
Approval	Offline email	**Online RA workflow**
Key generation	Browser-dependent	**Server-side, RSA 4096**
Automation	None	**EST · API**
CRL	Manual	**Automatic**

> ⚠ **Warning**
> Old CA certificates will **no longer be recognized after end of 2026** — about 45 days after this school. Migrate or re-apply **now**.

2 · Before You Start

Preparation checklist

IHEP Account / LDAP login works

You will log in to the CA WebUI with your usual IHEP username and password (IHEP Account, formerly IHEP SSO). Test it at login.ihep.ac.cn first.

Your LDAP email is correct

It becomes the certificate's email SAN, and the system uses it for ownership and notifications. Wrong email → lost certificates.

Decide your English name spelling

The certificate Subject needs ASCII letters only (space, dot, hyphen, apostrophe allowed). Use the same spelling forever — renewals must match.

Prepare an export password

A secret only you know, used to encrypt the downloaded `.p12` file. It is **not** your LDAP password and cannot be recovered by the CA.

Note

Open **https://gridca.ihep.ac.cn/webui/ihepca/** in a modern browser (Firefox recommended). The WebUI, EST and login all require HTTPS.

Three certificate types

Type	Purpose	Private key generated & held by	How to obtain
User	Grid user authentication — browser / CLI clients	CA generates; you download `.p12` and protect it	WebUI
Robot	Non-human identities: experiment services, pilots, automation	CA generates; the responsible person downloads `.p12`	WebUI
Host	Server authentication for machines	CA generates (WebUI first issue); **EST auto-renews on the host**	WebUI + EST

> **Note**
> Not sure which one you need? As a student doing analysis: **User Certificate**. Running a Belle II pilot service or an automated production task: ask your group leader about a **Robot Certificate**.

What is inside a certificate

Information	Where it lives	Can you change it?
Subject DN (name, account, OU, O)	The certificate itself	No — keep identical across renewals
SAN (email / DNS names)	The certificate itself	No — changed DNS or ownership ⇒ new application
Issuer · CRL URL · Certificate Policies	The certificate itself	No
Requester · Owner email · Comment	OpenXPKI system metadata	Only via workflow / RA

Your real certificate, as openssl shows it:

```
subject=C=CN, O=HEP, O=IHEP, OU=CC, CN=hanx14, CN=Xiao Han
issuer=C=CN, O=HEP, CN=Institute of High Energy Physics Certification Authority 2026
```

Note
In slash form — used by VOMS and older grid tools — your DN is: `/C=CN/O=HEP/O=IHEP/OU=CC/CN=hanx14/CN=Xiao Han`. Grid services map it to VOMS roles and DIRAC accounts: **keep your name spelling forever**.

3 · Hands-on: User Certificate

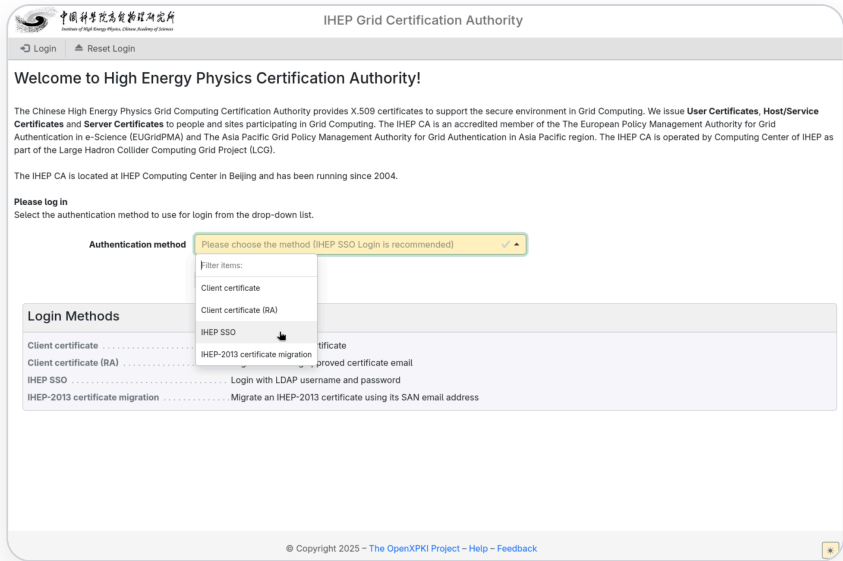
Step 1 — Log in with IHEP Account

1. **Open the WebUI**
   https://gridca.ihep.ac.cn/webui/ihepca/

2. **Choose authentication method**
   In the dropdown select **IHEP Account**.

3. **Enter IHEP credentials**
   Username + password, then **Login**.

4. **Verify your account**
   Top-right corner must show **your** account. Logout via the button — not by closing the tab.



The WebUI login page — pick **IHEP Account** in the Authentication method dropdown.

> ⚠ **Warning**
> `Login with the given credentials failed!` → check LDAP account/password, do **not** brute-force different accounts; contact CA support to check the LDAP mapping.

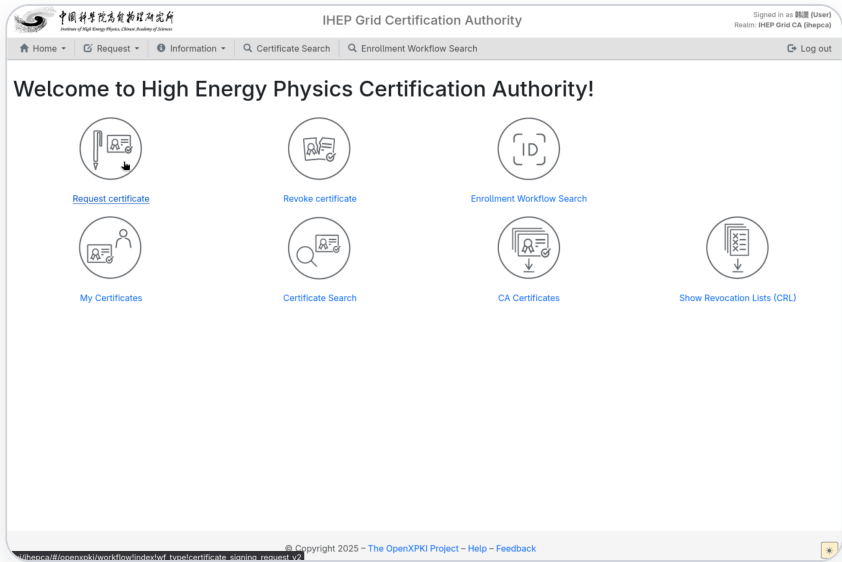
Step 2 — Know your two home pages

Home → My Certificates

- All certificates owned by your account
- Renewal and revocation start from here
- Download actions live in the certificate detail page

Home → My Workflows

- Every request you created
- Status, history, and error messages
- Bookmark the workflow ID before contacting support



After login — the top menu with **Request Certificate**.

> **Note**
> These two pages answer 90% of "where is my certificate" questions. If a certificate is not in **My Certificates**, check which account (email) you logged in with.

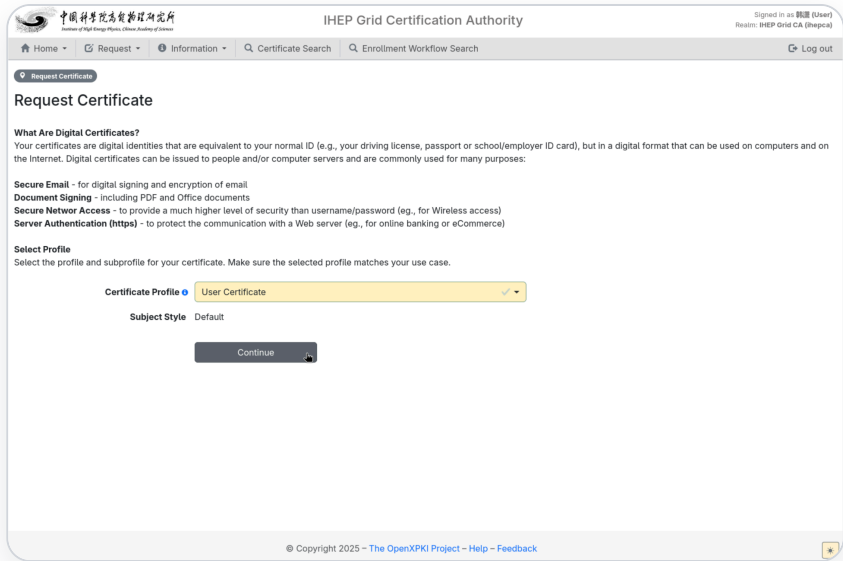
Step 3 — Start a request

1. **Top menu → Request Certificate** (**Request → Request Certificate** in older menus.)

2. **Select Profile: User Certificate**
   It is the default — click **Continue**.

The other profiles in the list:

- **Robot Certificate** — for experiment services and pilots (§7)
- **Host Certificate** — for machines (§7)



Select Profile — User Certificate is the default.

> **Note**
> **Continue** only saves and moves to the next page. **Submit** is what actually sends the request to the system and RA. Review the summary before every Submit.

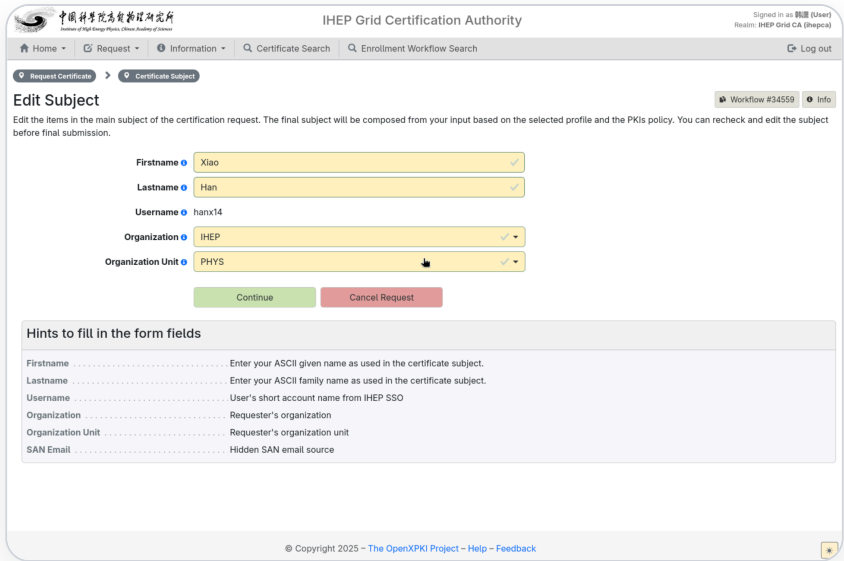
Step 4 — Edit Subject

1. **First name / Last name**
   ASCII only: letters, space, dot, hyphen, apostrophe. If LDAP returns your Chinese name, **type your English name yourself**.

2. **Username**
   Pre-filled from LDAP, locked — it binds the cert to your account.

3. **Organization Unit / Organization**
   Dropdowns — select your real department and institute.

4. **SAN Email**
   Provided by LDAP — leave it untouched.



Edit Subject — type your ASCII name; Username and email come from LDAP.

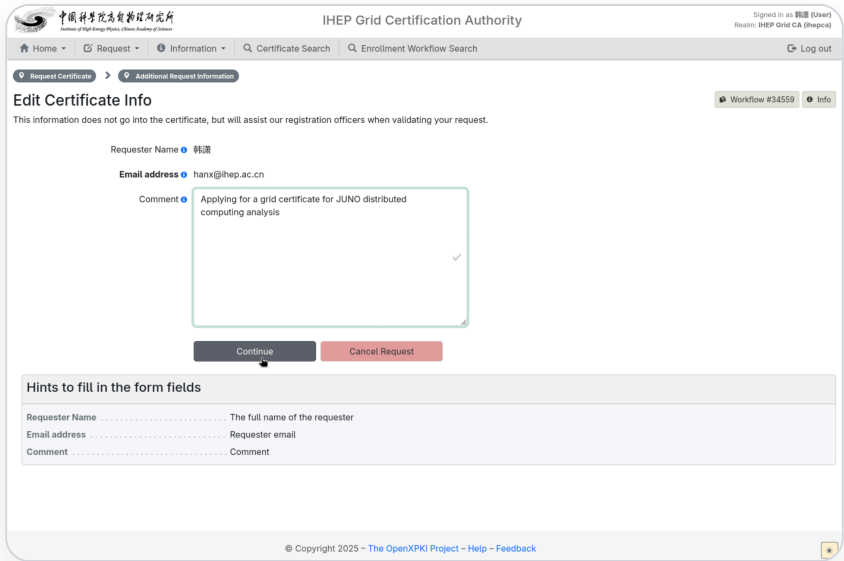
Step 5 — Certificate info

1. **Email address**
   Your LDAP email — the ownership and notification address.

2. **Requester**
   Your name, as the applicant.

3. **Comment**
   e.g. "Belle II analysis user, winter school". **Never** put passwords or private key info here. Then **Continue**.

> **Note**
> Nothing here changes your identity — identity fields come from LDAP. Comment is free text for the RA reviewers.



Edit Certificate Info — write your application reason in Comment.

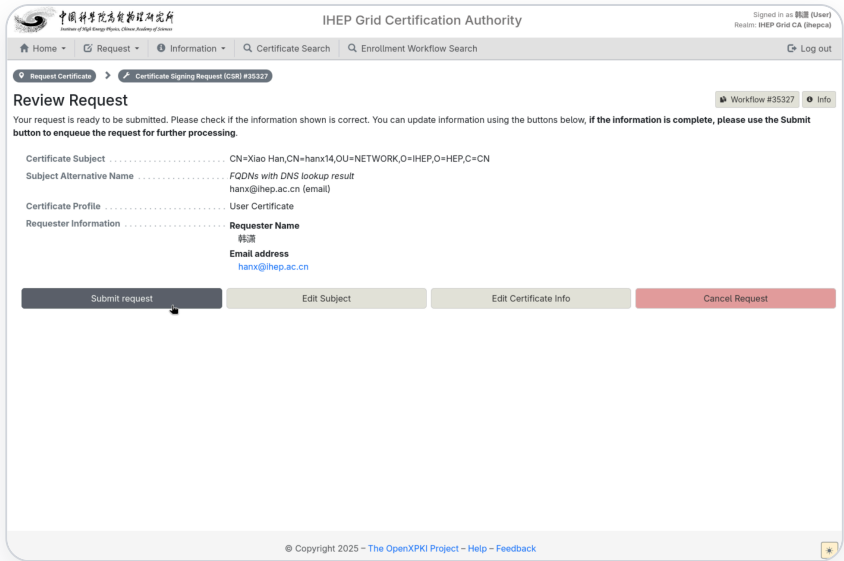
Step 6 — Review and submit

1 **Review Request**
check **Certificate Subject**, **SAN**, **Profile**, **Requester Information**; use **Edit Subject / Edit Certificate Info** to go back and fix anything.

2 **Submit request**

3 **Duplicate warning?**
Failed Policy Check means a repeated application — click **Submit with Policy Exception** and justify it (e.g. lost private-key password).

> **Note**
> Policy-exception requests show **Policy Violation Pending** and may take a little longer to be reviewed.



Review Request — the last chance to fix mistakes before Submit.

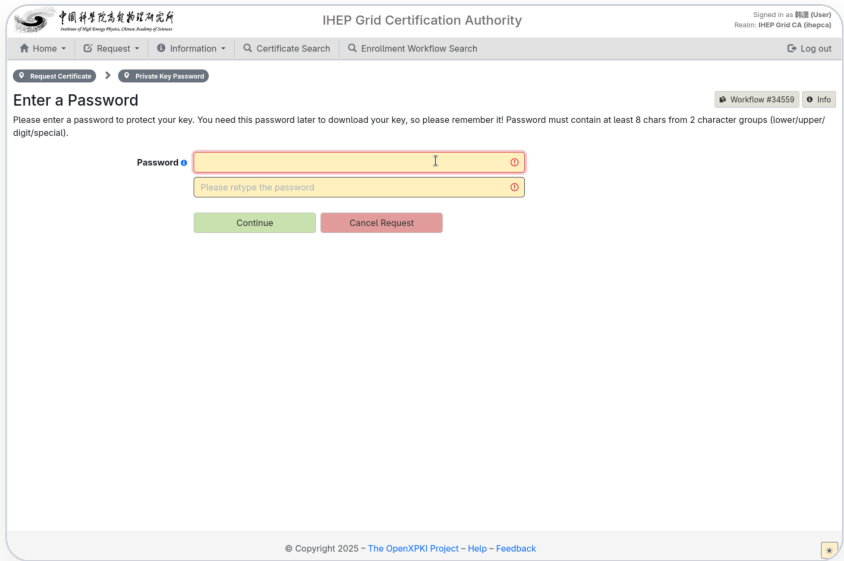
Step 7 — Set the private key password

1 **This is the Private Key Password**
It protects your private key — and you will type it **again when downloading** the certificate.

2 **Rules**
At least 8 characters, mixing at least two of: upper case, lower case, digits, special characters.

3 **Memorize it**
The CA cannot recover it. Losing it means re-applying via policy exception.

> **Note**
> Two different passwords exist: this **Private Key Password** (now), and an **Export Password** you will set when downloading the `.p12` (leave it empty to reuse this one). Neither is your LDAP password.

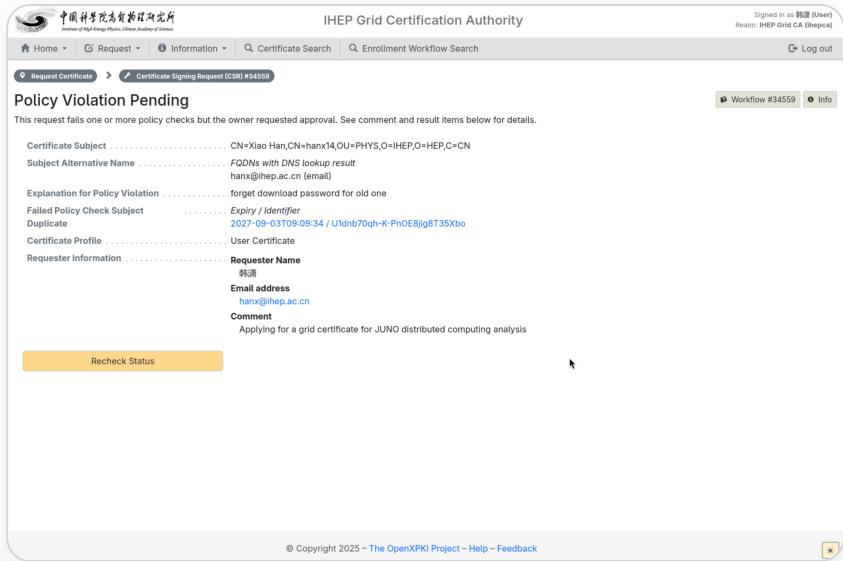


Enter a Password — the Private Key Password.

Step 8 — Track your request

Home → My Workflows — click the workflow ID to see State, History and messages.

Status	Meaning	Your action
`PENDING` / `ONHOLD`	Awaiting RA review	Wait; do **not** duplicate requests
`Policy Violation Pending`	Policy exception under review	Use **Recheck Status** to refresh
`SUCCESS`	Issued	Download from My Certificates
`REJECTED`	RA refused	Read the reason, fix, re-apply
`FAILURE`	Workflow error	Save the workflow ID, contact support



After submission — waiting for RA review (here: a policy-exception request).

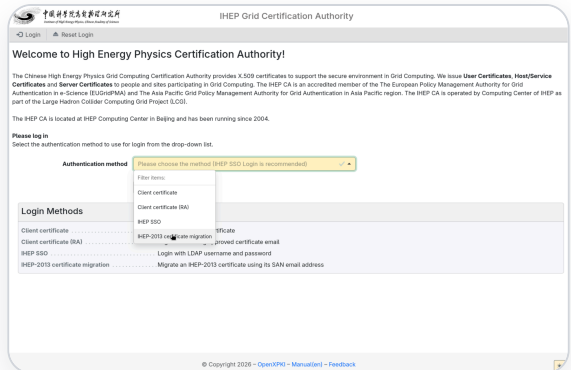
> ⚠ **Warning**
> The browser Back button does **not** cancel a request. Use the on-page **Cancel** action if you truly need to withdraw one.

4 · Migrate from IHEP-2013

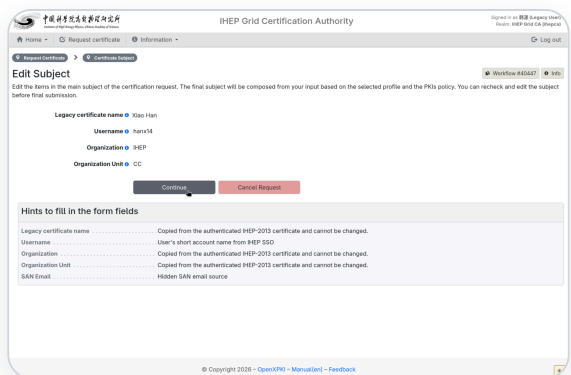
Already have an old personal certificate?

Migration entry = one-time shortcut for users holding a valid, unrevoked IHEP-2013 **personal** certificate:

1. Install the old personal certificate **with its private key** into your browser first.
2. Open the WebUI — the browser pops up the **certificate selection**: pick your **personal** IHEP-2013 certificate.
3. In **Authentication method** choose **IHEP-2013 certificate migration** and click **Login**.
4. Identity fields are **auto-filled** from the old cert + LDAP — no re-typing. Set a password, submit, download.



The login page — choose **IHEP-2013 certificate migration**.



Subject **auto-filled** from the old certificate and LDAP — read-only.

5 · Use It on the Grid

Download your PKCS#12

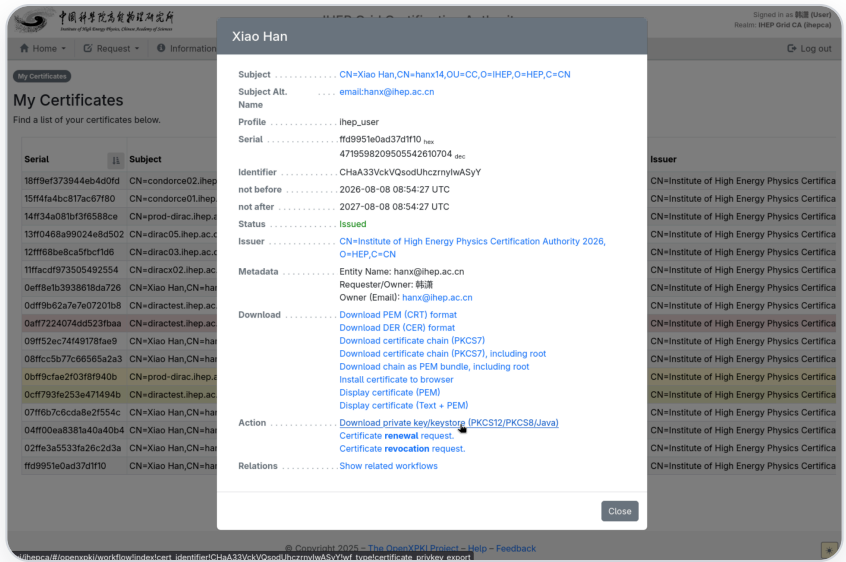
1. **Issued notification email**
   arrives with a direct link that opens your certificate.

2. **Home → My Certificates**
   click the certificate's Subject or serial number.

3. **Action → Download private key/keystore**

4. **Move the `.p12` somewhere safe**
   Out of the shared Downloads folder, restricted permissions. Not in /tmp, not on cloud drives.

> **Note**
> No download action? Either the private key is not held by the CA anymore, or the certificate came from EST — it does not mean the certificate is invalid.



Certificate detail — the **Action** area with the download entry.

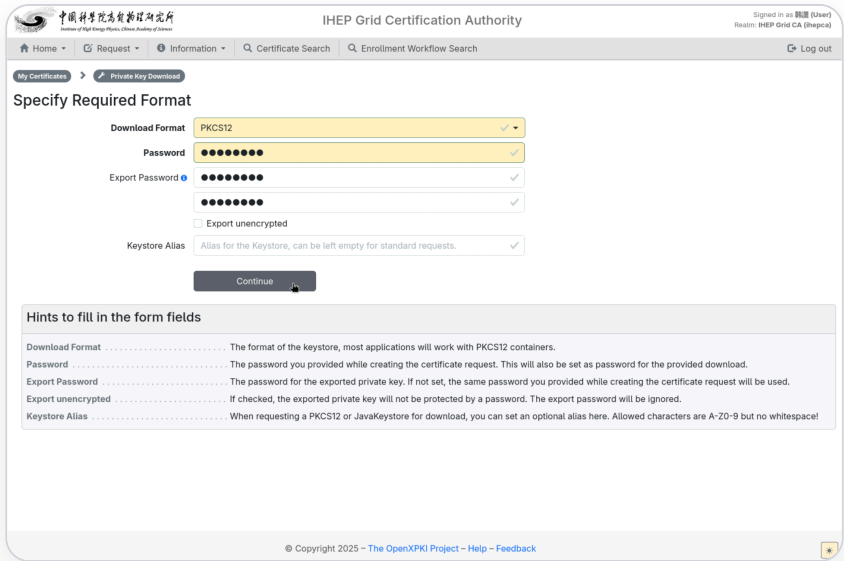
Download: format and the two passwords

1. **Download Format: PKCS12**
   confirm it on the **Specify Required Format** page.

2. **Password**
   enter the **Private Key Password** you set in Step 7.

3. **Export Password**
   encrypts the downloaded `.p12` — leave empty to reuse the private key password.

4. **Continue → SUCCESS → Download**
   save the file **immediately**.



Specify Required Format — PKCS12, Private Key Password, Export Password.

> ⚠ **Warning**
> The download entry **disappears once you leave the SUCCESS page**. If you miss it, reopen the certificate and start the download action again.

Install into Firefox

1. **Settings → Privacy & Security → Certificates → View Certificates**

2. **Your Certificates → Import**
   pick your `.p12`, enter the export password.

3. **Verify the entry**
   double-click it: **Issued By** = new IHEP Issuing CA; SAN contains your email.

4. **Try a client-cert website**
   pick your new certificate when the browser asks.



Firefox **Certificate Manager → Your Certificates → Import** — pick your `.p12` and enter the Export Password.

Important
Importing only a `.crt` / `.pem` (public part) does **not** work — the browser needs the private key, which lives in the `.p12`.

Tip: stick to one browser through apply → download → import, and avoid browser updates mid-workflow.

Export PEM files for the grid CLI

Rename the downloaded `.p12` — it is usually named after you, e.g. `Xiao Han.p12` — to `myCert.p12`, and upload it to **KEKCC or whichever server you will use** (e.g. with `scp`). Grid tools expect `~/.globus/usercert.pem` + `userkey.pem`:

```
mkdir -p "$HOME/.globus" && chmod 700 "$HOME/.globus"

openssl pkcs12 -in myCert.p12 -clcerts -nokeys -out "$HOME/.globus/usercert.pem"
openssl pkcs12 -in myCert.p12 -nocerts -out "$HOME/.globus/userkey.pem"
chmod 644 "$HOME/.globus/usercert.pem"
chmod 600 "$HOME/.globus/userkey.pem"
```

Verify what you got (subject, issuer, dates, SAN, and key match):

```
openssl x509 -in "$HOME/.globus/usercert.pem" -noout -subject -issuer -dates -ext subjectAltName
test "$(openssl x509 -in "$HOME/.globus/usercert.pem" -noout -modulus | openssl sha256)" \
  = "$(openssl rsa -in "$HOME/.globus/userkey.pem" -noout -modulus | openssl sha256)" \
  && echo 'certificate and key match'
```

> ⚠ **Warning**
> Delete `myCert.p12` on that server once the conversion succeeds — don't leave copies of the encrypted key around. If `openssl pkcs12` complains about the format, drop or add `-legacy`. Never send the `.p12` to anyone to "fix" it.

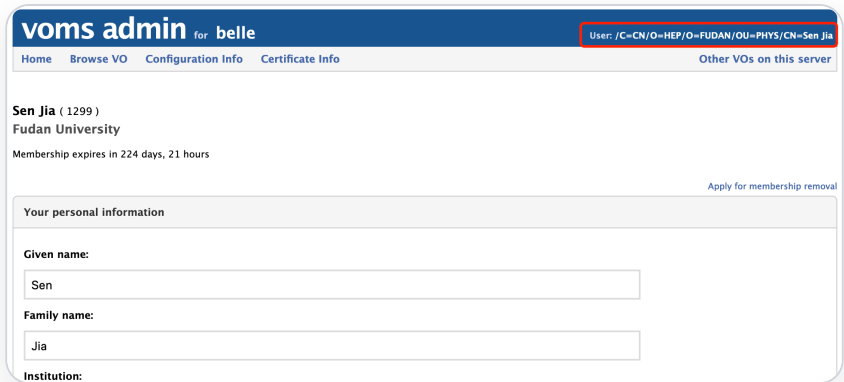
Register your Belle II VO membership

1. **Open the Belle II VOMS page**
   voms.cc.kek.jp:8443/voms/belle — log in **with your new certificate** (your first real use of it!).

2. **Request membership**
   follow the page actions to register as a new member.

3. **Wait for approval**
   a VO admin confirms — usually within a day or two.

4. **Done**
   your DN now carries the `/belle` roles used by grid services.



VOMS Admin for the belle VO — log in with your certificate; your DN and membership appear on your profile page.

> ⚠ **Warning**
> Membership binds to your **DN**. A normal renewal keeps your DN — membership survives. But if your DN ever changes (different name spelling), you must register again.

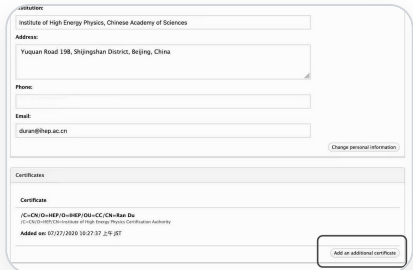
New certificate: update your VOMS binding

1. **Log in with the old certificate**
   the one already bound to your VO account.

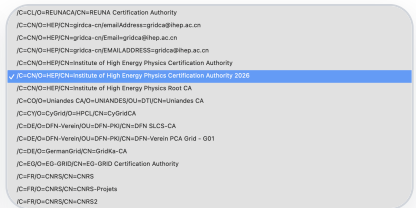
2. **Add an additional certificate**
   on your profile page, under **Certificates**.

3. **Select the new issuer**
   Institute of High Energy Physics Certification Authority 2026 in the CA list.

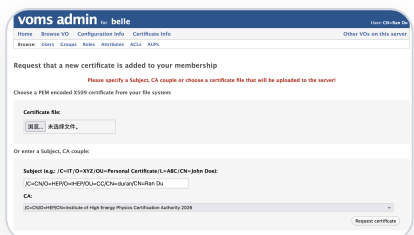
4. **Enter your new DN**
   slash form, e.g.
   `/C=CN/O=HEP/O=IHEP/OU=CC/CN=duran/CN=Ran Du` → **Request certificate**, then wait for VO admin approval.



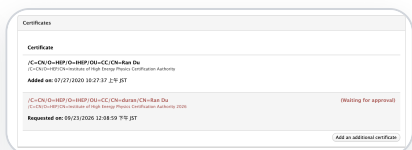
Profile → **Add an additional certificate**



Pick the **2026 issuer** in the CA list



Enter the new DN — or upload the **PEM only**



New certificate **waiting for approval**

Important
The form also accepts a **certificate file upload** — if you use it, upload the **PEM certificate only** (`usercert.pem`). **Never upload your `.p12` — it contains your private key.**

Use it: gBASF2 proxy example

On an IHEP login node (or KEKCC):

```
source /cvmfs/belle.kek.jp/grid/gbasf2/pro/bashrc
gb2_proxy_init
```

What the proxy output shows:

```
subject      : /C=CN/O=HEP/O=IHEP/OU=CC/CN=hanx14/CN=Xiao Han
issuer       : /C=CN/O=HEP/CN=Institute of High Energy Physics
               Certification Authority 2026
timeleft     : 23:53:58
DIRAC group  : belle
VOMS         : True
VOMS fqan    : ['/belle']
```

> **Note**
> The proxy is a **short-lived credential** derived from your certificate. Your long-lived secret (private key) never leaves your machine.

Remember:

- Certificate expires in ~1 year — renew in time (§6)
- New certificate ⇒ new proxy: re-run `gb2_proxy_init`
- Belle II VO membership (VOMS `/belle`) is registered against your DN — changing your DN breaks it

gbasf2: setup and your first job

Setup (KEKCC or IHEP login node):

```
source /cvmfs/belle.kek.jp/grid/gbasf2/pro/bashrc
gb2_check_release   # check your gbasf2 release
```

- ⓘ **basf2 and gbasf2 are NOT compatible** — use separate shell sessions
- gbasf2 and basf2 versions are **independent** of each other

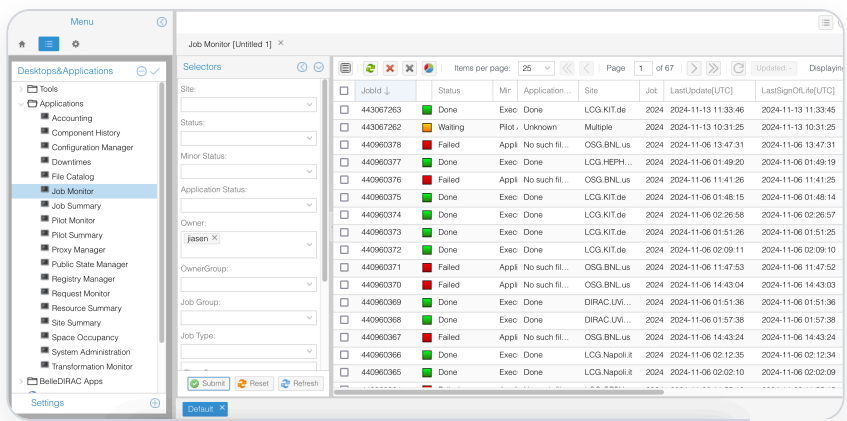
Golden rule

> **Important**
> Always run your script with **basf2 first** on local samples — only then submit it to the grid.

Submit a job — same logic as basf2:

```
gbasf2 ./ana01.py -s release-08-01-05 \
  -p 20261114_myproject --input_dslist input1.txt
```

- `-s` basf2 release · `-p` your project name · `--input_dslist` input dataset
- Monitor jobs in the **DIRAC portal**



The BelleDIRAC **Job Monitor**

Useful commands and where to get help

Command	Purpose
`gb2_job_status -p PROJ`	Job status of your project
`gb2_job_output -p PROJ --status Failed`	Fetch logs of failed jobs
`gb2_job_reschedule -p PROJ`	Retry failed jobs
`gb2_job_kill -p PROJ`	Kill remaining jobs
`gb2_ds_list /belle/user/<you>/PROJ`	Count output files
`gb2_ds_get /belle/user/<you>/PROJ`	Download outputs
`gb2_ds_rm "/belle/user/<you>/*"`	Remove obsolete datasets

Questions & discussion

- questions.belle2.org — the computing Q&A forum

Mailing list

- comp-users-forum@belle2.org — user discussion; subscribe at the list page

Documentation

- Belle II computing documentation — online book (BASF2 / gbasf2 / grid)

> **Note**
> For **certificate** problems (login, issuance, renewal, revocation) — contact the IHEP CA support, not the Belle II forum.

6 · Renewal & Revocation

Renew before expiry

1. **My Certificates → open a still-valid certificate**

2. **Action → Renew**
   profile and Subject are carried over — verify, don't retype.

3. **New export password → Submit**

4. **Download the new `.p12`, reinstall**
   browser + `.globus`, then make a fresh proxy.

Good to know:

- Renewal issues a **new private key and serial** — never stitch the new cert onto the old key by hand
- The old certificate stays valid until its original expiry → smooth switchover
- Renew ≠ change: different department, name spelling or ownership = **new application**, not renewal
- Expiry reminder emails are sent automatically — keep your LDAP email correct

Revoke when compromised

When to revoke immediately:

- Private key or `.p12` suspected leaked
- You stop being the responsible person (Robot/Host)

How:

1. **My Certificates → open the certificate**
2. **Action → Revoke**
   choose the most accurate reason, explain for the RA.
3. **Submit**
   and wait for CRL publication.

> **Important**
> **Revocation is irreversible.** If the key may be compromised: stop using it first, then revoke — do not wait for expiry, and do not just delete local files (copies may exist elsewhere).

Propagation: relying parties fetch the CRL on their own schedule — a revoked cert may still be accepted for a while by some services.

7 · Robot & Host Certificates

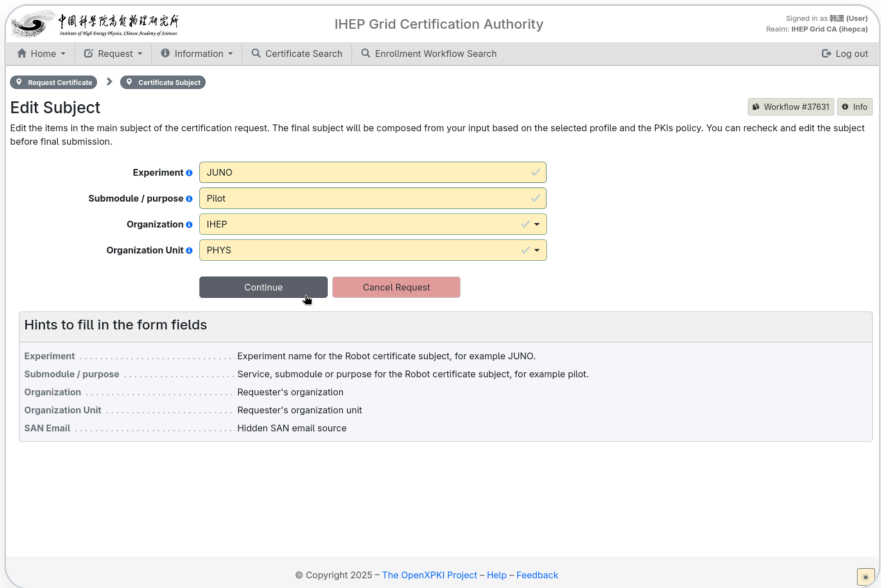
Robot certificates

For **non-human identities**: experiment pilots, production automation, service agents — not for natural persons.

Application highlights (WebUI, like §3):

- **Experiment** — e.g. `Belle` (letters, digits, dot, underscore, hyphen)
- **Submodule / purpose** — e.g. `pilot`
- **OU / O** — the responsible group
- SAN email = the responsible person's LDAP email (automatic)

> **Important**
> Every Robot certificate needs a **responsible person**. Ownership change, purpose change or key leak ⇒ revoke and re-apply — do not "migrate" a Robot cert to someone else.



Robot certificate — Experiment + Submodule / purpose; the second CN is fixed to **Robot**.

Resulting DN shape:

```
CN=Belle pilot,CN=Robot,OU=<dept>,O=<inst>,O=HEP,C=CN
```

Host certificates & EST

WebUI first issue (site admin):

- **Hostname** = primary FQDN (`host01.ihep.ac.cn`, never short names)
- **Additional hostnames** = one FQDN per entry, no wildcards, no IPs/URLs
- Names are issued lowercase; email SAN = applicant's LDAP mail

Install to the host (`.p12` → PEM):

```
openssl pkcs12 -in host01.p12 -clcerts -nokeys \
  -out /etc/grid-security/hostcert.pem
openssl pkcs12 -in host01.p12 -nocerts -nodes \
  -out /etc/grid-security/hostkey.pem
chmod 444 /etc/grid-security/hostcert.pem
chmod 400 /etc/grid-security/hostkey.pem
```

EST auto-renewal (hands-free afterwards):

- Script `ihep-host-auto.sh` on the host
- Renews automatically within **45 days** of expiry
- Old IHEP-2013 host certs migrate through the same script (mTLS, one-time per hostname)
- Site admins deploy it — users normally never touch this

> **Note**
> Full details (wrapper script, cron, RELOAD_CMD) in the **IHEPCA User Guide** — linked at the end.

8 · Troubleshooting & Security

Common problems

Symptom	Check first
WebUI login fails	LDAP account/password; wrong client cert selected; system clock
Cannot find my certificate	**My Certificates** + which email you logged in with
Stuck in `PENDING`	workflow ID from **My Workflows** → contact RA; don't re-submit
No "download key" action	Key not CA-generated / already exported / EST-issued — not an error
Browser never offers the certificate	Import the `.p12` (with key!); check cached choices
Certificate chain fails	Client must trust the **new Root + Issuing CA**; CRL URLs reachable

Note

Contact support with: certificate serial or Subject · workflow ID · time (UTC) · full error text. **Never** send private keys, `.p12`, or passwords.

Security habits

Move `.p12` out of Downloads immediately

Restricted location and permissions. Not /tmp, not sync drives, not group-readable home.

One export password, kept secret

Not the LDAP password; never recovered by the CA; never shared.

Revoke on suspicion

Lost key or leaked file → stop using it and revoke. Fast beats perfect.

Keep your LDAP email current

Expiry alerts and ownership follow that address.

Never

e-mail / chat / cloud-share a `.p12`, `userkey.pem` or password

Never

write passwords or key material in the Comment field

Summary — do it this week

1. **Check your IHEP Account login and LDAP email**
   — everything hangs on these two.

2. **Request / migrate your certificate**
   at gridca.ihep.ac.cn — new request, or the IHEP-2013 migration entry.

3. **Install to browser + `.globus`**
   — bring a working proxy to the hands-on session.

4. **Register / update your VOMS membership**
   join the belle VO — or add your new DN.

Important

Deadline: old-CA certificates stop being recognized at the **end of 2026**.

IHEP CA WebUI
gridca.ihep.ac.cn/webui/ihepca

IHEP CA user guide
cchelp.ihep.ac.cn/grid/new-ca

gbasf2 user manual
Belle II online book — gbasf2

Belle II VOMS
voms.cc.kek.jp/voms/belle

Questions & forum
questions.belle2.org

DIRAC portal
dirac.cc.kek.jp/DIRAC

IHEP Computing Center — PKI Team

Reporter	Xiao Han · *IHEP, CC*
Current Developer	Xiao Han · *IHEP, CC*
People	Xiao Han · *CA Administrator* · *RA Administrator* Gongxing Sun · *RA Administrator (former)* · Tian Yan · *CA Administrator (former)*
Acknowledgements	The Interoperable Global Trust Federation (IGTF) · The Asia Pacific Grid Policy Management Authority (APGridPMA) · The European Policy Management Authority for Grid Authentication (EUGridPMA)
WebUI	gridca.ihep.ac.cn

Questions?

Thank you!